Virtual University of Pakistan

BT501

Health Biotechnology

Midterms Past Papers

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MID TERM PAST PAPERS

Solved by Areha Maham and Zareen Fatima

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What is CFTR..? 2

Ans: • Cystic fibrosis (CF, MIM# 219700) is an autosomal recessive disorder of epithelial ion transport caused by mutations in the CF transmembrane conductance regulator gene (CFTR).

• CFTR is a cAMP-regulated chloride channel that regulates other ion channels. CFTR maintains the hydration of secretions within airways and ducts through the transport of chloride and inhibition of sodium uptake. • Dysfunction of CFTR can affect many different organs, particularly those that secrete mucus, including the upper and lower respiratory tracts, pancreas, biliary system, male genitalia, intestine, and sweat glands.

Techniques in rDNA Technology..? 2

Ans: • Gel electrophoresis • Cloning libraries • Restriction enzyme mapping • PCR • Nucleic Acid Hybridization • DNA Microarrays

Differnce b/w allogenic and autologus stem cells? 3

Ans:

Autologous Stem Cells: Sources of the patient's own stem cells (autologous) are either the cells from patient's own body or his or her cord blood. For autologous transplants physicians now usually collect stem cells from the peripheral blood rather than the marrow	Allogenic Stem Cells: Sources of stem cells from another donor (allogeneic) • are primarily relatives (familial-allogeneic) or completely unrelated donors (unrelated- allogeneic) • The stem cells in this situation are extracted from either the donor's body or cord blood

Short note on Alzheimer disease.? 3

Ans: Alzheimer diseases (AD) is a panethnic, genetically heterogeneous disease; less than 5% of patients have early-onset familial disease, 15% to 25% have late-onset familial disease, and 75% have sporadic disease.

• Approximately 10% of familial AD exhibits autosomal dominant inheritance; the remainder exhibits multifactorial inheritance.

• Current evidence suggests that defects of β-amyloid precursor protein metabolism cause the neuronal dysfunction and death observed with AD.
• Consistent with this hypothesis, mutations associated with early-onset autosomal dominant AD have been identified in the β-amyloid precursor protein gene (APP), the presenilin 1 gene (PSEN1), and the presenilin 2 gene (PSEN2).

AD is a central neurodegenerative disorder, especially of cholinergic neurons of the hippocampus, neocortical association area, and other limbic structures. • Neuropathological changes include cortical atrophy, extracellular neuritic plaques, intraneuronal neurofibrillary tangles, and amyloid deposits in the walls of cerebral arteries. • The neuritic plaques contain many different proteins including Aβ42/43 and apolipoprotein E. The neurofibrillary tangles are composed predominantly of hyperphosphorylated tau protein; tau helps maintain neuronal integrity, axonal transport, and axonal polarity by promoting the assembly and stability of microtubules.
Phenotype and Natural History
• AD is characterized by a progressive loss of cognitive function including recent memory, abstract reasoning, concentration, language, visual perception, and visual-spatial function. • Beginning with a subtle failure of memory, AD is often attributed initially to benign "forgetfulness." Some patients perceive their cognitive decline and become frustrated and anxious, whereas others are unaware. • Eventually, patients are unable to work, and they require supervision. Social etiquette and superficial conversation are often retained surprisingly well.
• Ultimately, most patients develop rigidity, mutism, and incontinence and are bedridden. Other symptoms associated with AD include agitation, social withdrawal, hallucinations, seizures, myoclonus, and parkinsonian features. • Death usually results from malnutrition, infection, or heart disease.

Diabetes as polygenic disorder.? 5

Ans: What is a polygenic disease
• Examples of polygenic conditions include hypertension, coronary heart disease, and diabetes. Because such disorders depend on the simultaneous presence of several genes, they are not inherited as simply as are singlegene diseases.

Genetic disorders may also be complex, multifactorial, or polygenic, meaning they are likely associated with the effects of multiple genes in combination with

lifestyles and environmental factors. Multifactorial disorders include heart disease and diabetes

Diabetes mellitus is the most common endocrine disease, consuming almost 15% of annual health care expenditures in the United States. The focus of this review is on type II (non-insulin-dependent) diabetes mellitus, which accounts for approximately 90% of diabetic patients. The current understanding of the pathogenesis of type II diabetes is multifaceted. Complex defects in both insulin action and insulin secretion produce the metabolic derangements responsible for the disease. The pathophysiology is the result of a combination of polygene defects and environmental factors. The search for the responsible gene or genes is described and several candidate genes are discussed. Diabetogenes are addressed in the context of type II diabetes representing a prototype for the identification of the genetic basis of common diseases.

Human genome project.? 5

International Human Genome Sequencing Consortium:
- Proposed 1985, endorsed in 1988.
- 20 governmental groups.
- "Public project." Craig Venter & Celera Genomics:
- Founded 1998.
- Sequence in 3 years.
- Technology: automation, computers.
- Had access to public project's data.

• Approach was conservative and methodical.
• Had to wait for technology.
• First produced a clone-based physical map of the genome that would serve as a scaffold for the later sequence data:
• Broke genome into chunks of DNA whose position on chromosome was known from maps, clone into bacteria using BACs.
• Digest BAC-inserted clonal chunks of DNA into small fragments.
• Sequence small fragments.
• Stitch together BAC clones to assemble sequence.
• Assemble genome sequence from BAC clone sequences, using clone-based physical map. **Celera**
• Approach using "shotgun sequencing" (no organized map).
• Shreds genome randomly into small fragments with no idea of where they are

physically located.
• Clones and sequences fragments.
• Uses computer to stitch together genome by matching overlapping ends of sequenced fragments.

Timeline

• Genome sequencing driven by technology.
• 1985: 500 base pairs per day by hand.
• 1985-86: PCR and automated DNA sequencing.
• 1992: BACs.
• 2000: 1000 bases per second.

Waiting for Technology

• Eyes on the human genome.
• While waiting for technology other genomes were sequenced. Current Status
• Human genome ~3.2 Gb.
• "Rough draft" sequence of the human genome.
• Have sequenced 90% of the 2.5 Gb of gene-rich (euchromatic) DNA.
• What is considered finished?
• Fewer than 1 base in 10,000 is incorrectly assigned.
• More than 95% of the euchromatic regions are assigned.
• Each gap is smaller than 150 kb.

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Different question

How real time Pcr is used for immunoassay infectious agent?

The Polymerase Chain Reaction (PCR) is a process for the amplification of specific fragments of DNA. Real-Time PCR a specialized technique that allows a PCR reaction to be visualized "in real time" as the reaction progresses. As we will see, Real-Time PCR allows us to measure minute amounts of DNA sequences in a sample!

This method also allows a rapid assessment of the presence of antibiotic resistance genes or gene mutations.
• Although this genetic approach is not always predictive of phenotypic resistances, in specific
situations it may help to optimize the therapeutic management of patients.
• Finally, an approach combining the detection of pathogens, their mechanisms of

antibiotic resistance, their virulence factors and bacterial load in clinical samples could lead to profound changes in the care of these infected patients.

In recent years, quantitative real-time PCR tests have been extensively developed in clinical microbiology laboratories for routine diagnosis of infectious diseases, particularly bacterial diseases.

• This molecular tool is well-suited for the rapid detection of bacteria directly in clinical specimens, allowing early, sensitive and specific laboratory confirmation of related diseases.

• It is particularly suitable for the diagnosis of infections caused by fastidious growth species, and the number of these pathogens has increased recently.

Autosomal recessive disorder?

Ans: Autosomal Recessive Disorders (AR)

• The affected person has 2 abnormal alleles , that is he is a homozygote

• Heterozygotes are carriers of the abnormal gene and not affected

• Most of us carry one or two recessive potentially detrimental alleles

• For common AR disorders such as beta thalassemia, population screening for healthy carriers can identify couples at risk who benefit from prospective genetic counseling

• This need not involve DNA studies

• Both parents of an affected person (dd) are heterozygous for the mutant gene (Nd)

• Each sibling of an affected person has a 25% chance of being affected and a 2/3 chance of being a carrier (2 Nd and one NN among the non-affecteds

X-linked recessive inheritance

• Heterozygous mothers have a 50% chance to transmit the gene to each son or daughter, the son will be affected (because he has only one X chromosome carrying an abnormal allele, while the daughter will be a carrier)

• Hemizygous affected males transmit the gene to all their daughters but to none of their sons, daughters become carriers

Mitochondrial inheritance

• Matrilineal mode of inheritance: only mother passes mitochondrial DNA to offspring

• Higher spontaneous mutations than nuclear DNA affects both males and females , but transmitted only through females

• Has a wide range of severity due to heteroplasmy

• Example: diabetes mellitus with sensorineuronal deafness
• Only affected females can pass the disease to their sons and daughters, affected males have normal offspring

Importance of defining the mode of single gene inheritance

• The risks of recurrence vary with the category of single gene disorder, for example a father affected by AD condition has a risk of 50% of passing the condition to his son and daughter in each pregnancy.
• A couple with a boy affected by an autosomal recessive condition has a 25% chance to have another affected child with each pregnancy
• Molecular diagnosis are important diagnostic tools for single gene disorders and they can also help in defining the mode of inheritance in cases when the same phenotype is inherited in different ways

Conclusions

• An affected with autosomal dominant condition usually has one affected parent
• An affected with an autosomal recessive condition may have affected brothers and sisters and normal parents who are consanguineous
• Boys are affected by X-linked recessive condition when they get the abnormal allele from their carrier mother
• DNA tests can diagnose a large number of single gene disorder.